

Here is the complete, fully updated Master Ledger for the PI Cream Project. This comprehensive docket consolidates every formulation tier, complexion module, scent variant, clinical protocol, regulatory requirement, and safety guideline established across our development process.

PI CREAM — MASTER PROJECT LEDGER

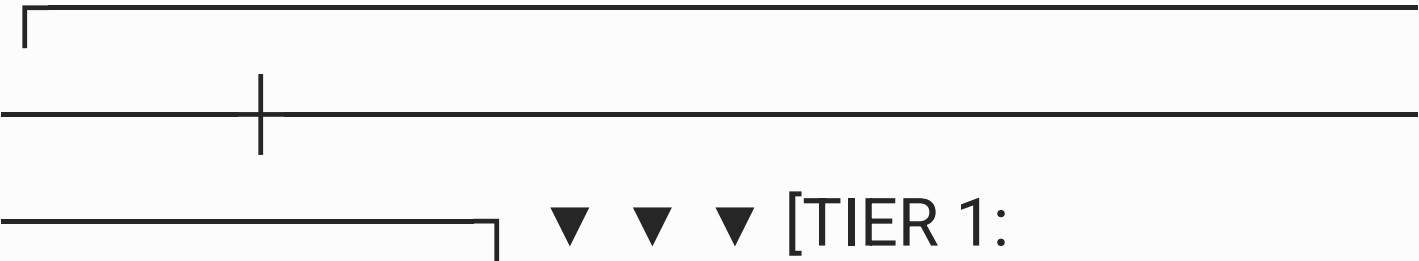
Project ID:PI-CREAM-

TRILOGY-2026Classification: Advanced Chrono-Dermal Phytoestrogenic Skincare Revision Date: July 2026 Target Audience: Unisex / High-Performance Anti-Aging & Post-Workout Recovery

1. Product Family Trilogy & Formulation Architecture

All three tiers are built on a Universal Bio-Active Base (~97%) consisting of a proprietary co-ferment (*Lactobacillus/Aloe Barbadensis Leaf/Kelp Ferment Filtrate*) combined with liposomal phytoestrogens, lipid-soluble Vitamin C (THD Ascorbate), and cold-pressed Sea Buckthorn Fruit Oil.

PI CREAM TRILOGY |



▼ ▼ ▼ [TIER 1:
PREVENTATIVE] [TIER 2: TRANSITIONAL] [TIER
3: REPLENISHING] Ages 35–45 Ages 45–55
Ages 55+ Hydro-Gel Cream Satin Cream-Gel
Cushion Velvet Cream

Master Formulation Specifications Matrix

Specifica tion	Tier 1: Preventat	Tier 2: Transitio	Tier 3: Replenish
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	ive (35–45)	nal (45–55)	ing (55+)
Primary Goal	Early Collagen Preservation & Hydration	Perimenopausal Density & Lipid Rebalancing	Postmenopausal Volume & Deep Barrier Repair
Aloe Vera / Kelp Co-Ferment	67.0%	60.0%	53.0%
Total Lipid Phase (Oils)	15.0% (Squalane, Rosehip, 0.5% Sea Buckthor	22.0% (Squalane, Rosehip, Pomegrate,	32.0% (Squalane, Rosehip, Pomegrate,

	n)	1.5% Sea Buckthor n)	3.0% Sea Buckthor n + Ceramide s)
Liposom al Phytoestr ogen Complex	0.5% Genistein & Red Clover	2.0% Genistein & Red Clover	3.0% Genistein & Red Clover <i>(Max ER-β Saturatio n)</i>
Turnover & Active Engine	1.0% Bakuchiol + 4% Niacinam ide + 2% THD	1.0% Bakuchiol + 3% Niacinam ide + 3% THD	Copper Tripeptid e-1 (GHK-Cu) + 5% THD Ascorbat

	Ascorbate	Ascorbate	e + Ceramide s NP/AP/ EOP
Sensory Finish	Weightless hydro- cream, satin finish	Silky cream- gel, velvety finish	Cushion- rich, deeply nourishing velvet cream

2. Complexion Bracketing System (Fitzpatrick Types I–VI)

To address how melanin density changes skin aging, UV vulnerability, and inflammatory responses, a Complexion Micro-Active (CMA) Module (~3%) is integrated into the universal base:

Complexion Bracket	Target Fitzpatrick Types	Primary Vulnerability	Complexion Micro-Module Additions
Fair / Pale	Types I – II	UV Photo-aging & Capillary Fragility	<i>Palmaria palmata</i> (Red Algae) + Ferulic Acid + Arnica Extract
Bronze / Tan	Types III – IV	Glycation & Mottled Hyperpigmentation	Carnosine (Anti-glycation) + 1.0% Alpha-Arbutin

Dark / Ebony	Types V – VI	Post- Inflamma tory Hyperpig mentatio n (PIH) & Ashy Dryness	2.0% Tranexa mic Acid + Acetyl Glucosa mine + Phytosph ingosine
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3. Male-Appealing Scent Variant Selection Matrix

Each variant relies on a 0.20% precision blend of 100% natural, cold-pressed essential oils added during Phase C (Cool-down below 35^{^{\circ}\text{C}}):

PI CREAM SCENT VARIANTS	

▼ ▼ ▼ [VARIANT A:
ALPINE FOREST] [VARIANT B: PACIFIC CEDAR]
[VARIANT C: SMOKED VETIVER] Siberian Fir,
Eucalyptus, Peppermint Cedarwood, Bergamot,
Black Pepper Haitian Vetiver, Cypress,
Cardamom Focus: Post-Workout Cooling Focus:
Daily AM Shield / Sebum Focus: PM Repair /
Post-Shave

Variant A (Alpine Forest): *Abies Sibirica*
(Siberian Fir) Oil, *Eucalyptus Globulus* Leaf
Oil, *Mentha Piperita* (Peppermint) Oil,
Tanacetum Annuum (Blue Tansy) Flower Oil.

Variant B (Pacific Cedar): *Cedrus Atlantica*
(Cedarwood) Bark Oil, *Citrus Aurantium*
Bergamia (Bergamot FCF) Fruit Oil, *Piper*
Nigrum (Black Pepper) Seed Oil, *Santalum*
Spicatum (Sandalwood) Oil.

Variant C (Smoked Vetiver): *Vetiveria*
Zizanioides (Vetiver) Root Oil, *Cupressus*

*Sempervirens (Cypress) Leaf Oil, Elettaria
Cardamomum (Cardamom) Seed Oil, Salvia
Sclarea (Clary Sage) Oil.*

4. Master INCI Ingredient List (IP-Fortified)

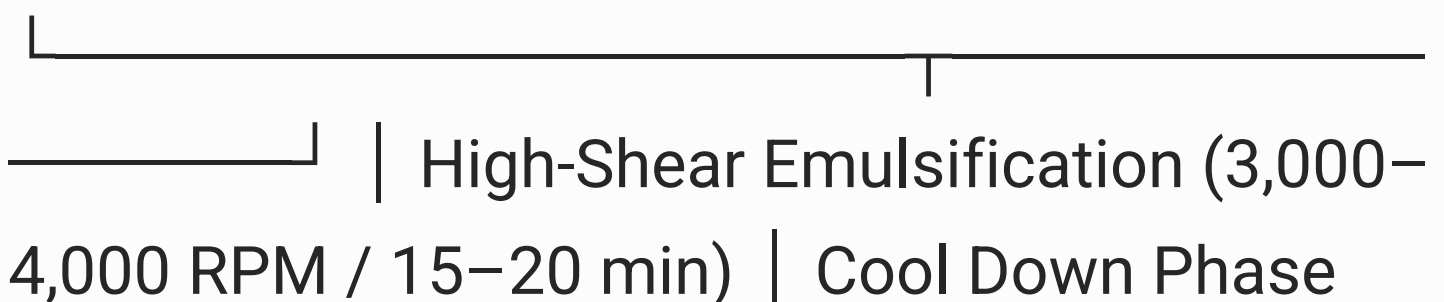
Universal Master INCI Deck: Lactobacillus/
Aloe Barbadensis Leaf/Kelp Ferment Filtrate,
Squalane, Rosa Canina (Rosehip) Seed Oil,
Punica Granatum (Pomegranate) Seed Oil,
Tetrahexyldecyl Ascorbate, Propanediol,
Glycerin, Cetearyl Alcohol, Glyceryl Stearate,
Hippophae Rhamnoides (Sea Buckthorn) Fruit
Oil, Panthenol, Soy Isoflavones (Genistein),
Trifolium Pratense (Red Clover) Flower
Extract, Niacinamide, Bakuchiol, Palmitoyl
Tripeptide-38, Copper Tripeptide-1, Ceramide
NP, Ceramide AP, Ceramide EOP,
Phytosphingosine, Cholesterol,

Phosphatidylcholine, Tocopherol, Sodium Hyaluronate, Cetearyl Glucoside, Lecithin, Polysorbate 80, Xanthan Gum, Carbomer, Sodium Lauroyl Lactylate, Sodium Phytate, Ethylhexylglycerin, Phenoxyethanol [+/- Fragrance (100% Natural Essential Oil Blend), *Limonene*, Linalool*, Citral*, Geraniol*].

**Declares natural essential oil variants A, B, or C and inherent terpene allergens per EU 1223/2009.*

5. Manufacturing Parameters & Phase Execution

[Phase A: Water & Co-Ferment Phase] [Phase B: Lipid & Oil Phase] Heat to 70°C–75°C Heat to 70°C–75°C



(Slow paddle mixing to $< 35^{\circ}\text{C}$) | [Phase C: Heat-Sensitive Actives & Scent] • Liposomal Genistein & Red Clover • THD Ascorbate (Lipid Vitamin C) • Peptides / Bakuchiol / Essential Oils • Eco-Cert Preservative System

Target pH Range: [5.00 - 5.50](#)

Viscosity Profile (25°C):

Tier [1: 15](#),000 - 25,000 cPs

Tier [2: 30](#),000 - 45,000 cPs

Tier [3: 55](#),000 - 75,000 cPs

QC Clearance: Microbial Total Plate Count < 100 CFU/g; Pathogens (*E. coli*, *S. aureus*, *P. aeruginosa*) ABSENT.

6. Chrono-Dermal Application Protocols

1. Post-Exercise Recovery Ritual (The 10-Minute Rule):

Allow body temperature to cool down until sweating stops ([5-10](#) \text{ min}).

Cleanse face with lukewarm water to remove sweat salts and urea.

Apply 1–2 pumps within 3 minutes while skin temperature (34°C – 36°C) and micro-circulation are naturally elevated for deeper active absorption.

2. Morning (AM Shield): Apply thin layer under broad-spectrum SPF 30+.
3. Night (PM Deep Repair): Apply 30 minutes before sleep to align with peak nighttime cell mitosis (10 \text{ PM}–2 \text{ AM}).

7. Safety, Allergens & Regulatory Warnings

Allergen Notice: Explicitly labels Soy Isoflavones (Genistein), Sea Kelp Ferment,

and Botanical Seed Oils on primary packaging.

Required Box Warning Text:WARNINGS:
FOR EXTERNAL USE ONLY. ALLERGEN
NOTICE: CONTAINS SOY-DERIVED
INGREDIENTS AND MARINE ALGAE
EXTRACTS. PERFORM A 24-HOUR
PATCH TEST PRIOR TO USE.
DISCONTINUE IF IRRITATION OCCURS.
AVOID DIRECT EYE CONTACT. KEEP
OUT OF REACH OF CHILDREN.

Compliance Approvals: Formulated in
accordance with US FDA MoCRA (Facility
Registration & Adverse Event contact) and
EU Regulation 1223/2009 (CPSR Parts A & B
+ CPNP Registration).

8. Updated Clinical Trial Protocol

Cohort Size: N = 90 Total Participants (N =

30 per tier cohort) across a 12-Week Double-Blind Study.

Instrumentation Suite:

50 MHz High-Frequency Ultrasound (HFUS): Measures Sub-Epidermal Low-Echogenic Band (SLEB) reduction and dermal density gains (\ge 18-25\%).

Cutometer® Dual MPA 580: Measures R_0 (firmness) and R_7 (biological elasticity gains of \ge 25-32\%).

Corneometer® CM 825: Confirms 48-hour hydration and TEWL barrier repair.

Antera 3D® Spectral Imaging: Quantifies R_z (wrinkle depth reduction up to 28\%) and melanin distribution.

Status: All formulation parameters, INCI decks, clinical testing specs, regulatory packaging copy, and manufacturing guidelines are locked in the master ledger

under PI CREAM TRILOGY.